# Daily Drilling Report
## SPD19B-03
## 3/20/2014

### Well Data

| Well Name | Operator | Field Name | Well Type 1 | Rig | Code |
|---|---|---|---|---|---|
| SPD19B-03 | Petropars Ltd. | South Pars Gas Field | Development | Deep Driller, DD6 | SP19-3B-01-DL-FOR-0... |
| **Report Number** | **Hole Section** | **Days LTI (days)** | **Water Depth (m)** | **Original KB Elevation (m)** | **Head Count** |
| 10.0 | 24" | 16.00 | 72.83 | 31.70 | 99.0 |
| **Last Cas** | **Daily Cost Total** | **Daily Mud Cost** | **Cum Cost To Date** | **AFE Number** | **Cum Time Log Days (days)** |
| 26" CP, 178.00mKB | 199,044.42 | 333.80 | 2,325,630.76 | | 8.60 |
| **Depth Start (mKB)** | **Depth End (mKB)** | **Depth End (TVD) (mKB)** | **Depth Progress (m)** | **Drilling Hours (hrs)** | **Average ROP (m/hr)** |
| 875.00 | 875.00 | 847.90 | 0.00 | | |

**Operations at Report Time**
CMT Job, in progress.

**Operations Summary**
Run back f/ 825m to Btm (875m). Sweep hole. POOH Rotary BHA to surface and R/B same. RIH 18 5/8" casing to 869.9 (Shoe depth). Land CHH on landing ring. Try to circulate, no success. Work to establish circulation, in progress.

**Operations next 24hrs**
Cont' work to establish circulation, perform CMT job , skid to well # 05.

### Supervisors Contact

| Job Contact | Position |
|---|---|
| Mojtaba Parvaresh | Drilling Supt. |
| Mohammad Ali Rezaei | Senior Drilling Sup. |
| Jabbar Mohammadi | Night Drilling Sup. |
| Hamed Daneshkhah | Drilling Engineer |
| Yaghub Mohammadi | Geologist |
| GH. R. Babaei | POGC Rep. |

### HSE Data

| Type | Last Date | Next Date |
|---|---|---|
| Abandon Drill | 3/9/2014 | 4/29/2014 |
| BOP Test | | |
| Chemical Spill Drill | | |
| Fire Drill | | |
| H2S Drill | | |
| Heli Crash Drill | | |
| Medical Drill | | |
| MOB Drill | 2/28/2014 | 4/30/2014 |
| Pit Drill | | |
| Trip Drill | | |

### STOP Cards

**Number of Reports:** 40

### Operations Summary

| Start Time | End Time | Dur (hrs) | Phase | Code 1 | Code 2 | Time P-N-T | Comment |
|---|---|---|---|---|---|---|---|
| 00:00 | 01:00 | 1.00 | 24" | D | 04 | P | Wash and R/D f/ 825m to 875m (Btm) w/ 400-600 gpm/ 500-750 psi/ 20-35 rpm/ 5-19 Kft-lbs. |
| 01:00 | 02:30 | 1.50 | 24" | D | 07 | P | Sweep hole by pumping 60 bbl GG + Circulate for 3 min + 100 bbl PHG w/ 1100 gpm/ 2100 psi/ 20 rpm while reciprocate the string. |
| 02:30 | 03:00 | 0.50 | 24" | D | 09 | P | Spot first batch of 10 ppg weighted PHG (340 bbl) f/ TD to expected top of Sachun formation (685m) w/ 600 gpm/ 680 psi, while reciprocating string w/ 20 rpm. |
| 03:00 | 03:30 | 0.50 | 24" | D | 04 | P | POOH Rotary BHA#3 f/ 875m to 685m. Hole in good condition. |
| 03:30 | 04:00 | 0.50 | 24" | D | 09 | P | Spot second batch of 10 ppg weighted PHG (340 bbl) to loss zone (495m) w/ 600 gpm/ 620 psi/ 20 rpm. |
| 04:00 | 07:30 | 3.50 | 24" | D | 04 | P | Cont' POOH f/ 685m to surface in good hole condition. L/D Jar. R/B BHA. Check the bit (1-1-WT-G-E-I-NO-TD). |
| 07:30 | 09:00 | 1.50 | 24" | G | 01 | P | L/D drilling bails, P/U and M/U fill up tool to TDS. Install 22' long bails. R/U AL-Ghaith side door elevator, power tong and single joint elevator.<br><br>Note: Use one 5 7/8" pup joint (5 ft) above ( X/O + Fill up tool) for adjusting the length. |
| 09:00 | 10:30 | 1.50 | 24" | G | 18 | P | HPJSM. P/U & M/U 18 5/8" (97 ppf, BTC, K-55) Shoe joint (Botil float shoe already made up) + 1 x Jt 18 5/8" Casing + Float Collar joint (Botil float collar already made up) and RIH to 35m.<br><br>*** Check Float shoe & Float collar for passing thru & holding, Ok.<br>*** Apply thread lock on shoe track connections.<br><br>Note: Al-Ghaith back up powe tong torque gauge was not working. |
| 10:30 | 12:30 | 2.00 | 24" | G | 18 | P | Cont' P/U & M/U 18 5/8" CSG and RIH f/ 35m to 178m (26" CP shoe). |
| 12:30 | 13:30 | 1.00 | 24" | G | 01 | P | Remove 18 5/8" hand slips and Side door elevator, R/U 18 5/8" Spider elevator and Spider slips. |
| 13:30 | 14:00 | 0.50 | 24" | G | 07 | P | Circulate 1.2 casing string volume w/ 450 gpm, 120 psi. |
| 14:00 | 21:00 | 7.00 | 24" | G | 18 | P | RIH 18 5/8" CSG and RIH f/ 178m to 847m.<br><br>*** Fill string w/ fill up tool every joint while RIH.<br>*** Install 18 5/8" x 24" bow centralizers as per tally. |

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

### Operations Summary

| Start Time | End Time | Dur (hrs) | Phase | Code 1 | Code 2 | Time P-N-T | Comment |
|---|---|---|---|---|---|---|---|
| 21:00 | 22:00 | 1.00 | 24" | G | 18 | P | P/U & M/U Aker 20 3/4" 3k CHH Assy. (1 x 18 5/8" nipple + 20 3/4" CHH + RT+ 2 x 18 5/8" pup jts) in one piece + 1 jt 18 5/8" CSG. Wash down w/ 200 gpm & 180 psi f/ 847m to 869.9m (Shoe Depth) and Land CHH on 26" landing ring. S/O casing weight completely, no movement of CP.<br><br>*** 18 5/8" Shoe @ 869.9m & Float collar @ 845.8m.<br>*** Totally used 73 joints of 18 5/8" Casing (K-55, 97 PPF, BTC) + 1 x 18 5/8" Nipple + 19 no of Botil 18 5/8" x 26" Bow centralizer + 1 x Botil 18 5/8" x 26 " Rigid centralizer + 20 no Stop collar. |
| 22:00 | 00:00 | 2.00 | 24" | G | 07 | P | M/U 18 5/8" Circulating Swage. Try to circulate, pressure increasing sharply and not possible to circulate. Check surface lines,ok.Try to establish circulation several times, no success. L/D Circulating Swage. P/U string and stab in fill up tool (P/U & S/O was normal, CSG free). Try to establish circulation while shaking up & down the string and keeping 1000psi inside CSG, no success.<br><br>Mar 21, 2014<br><br>00:00 - 02:00 Cont' shaking up & down the string while keeping 1000psi inside CSG, establish circulation.<br><br>02:00 - 02:30 Land CHH again on landing ring, Ok.<br><br>02:30 - 04:00 M/U Swage and circulate 1.2 casing string volume w/ 450 gpm & 200psi.<br><br>04:00 - 05:00 B/O circulation swage. M/U cementing X/O (18 5/8" BTC x Quick Union). Drop bottom plug. M/U Mehran Single cement head. (Top plug already loaded). Line up surface line to cement unit.<br><br>05:00 - 06:00 HPJSM. Flush and pressure test surface line to 500 / 2000 psi for 5/10 min, Ok. Mehran CMT unit start pumping Lead CMT slurry 12.5 ppg w/ 7-8 bpm, 350 - 420 psi, in progress. |

### Mud Information

| Depth (mKB) | Type | Density (lb/gal) | Funnel Viscosity (s/qt) | PV Calc (cp) | YP Calc (lbf/100ft²) | Filtrate (mL/30min) |
|---|---|---|---|---|---|---|
| 875.00 | Sea Water | 8.60 | | | | |
| **pH** | **Pf (mL/mL)** | **Mf (mL/mL)** | **Pm (mL/mL)** | **Sand (%)** | **Low Gravity Solids (%)** | **High Gravity Solids (%)** |
| | | | | | | |
| **Vis 3rpm** | **Vis 6rpm** | **Percent Oil (%)** | **Percent Water (%)** | **Solids (%)** | **Chlorides (mg/L)** | **Filter Cake (/32")** |
| | | | | | | |
| **Gel 10 sec (lbf/100ft²)** | **Gel 10 min (lbf/100ft²)** | **Gel 30 min (lbf/100ft²)** | **Oil Water Ratio** | **T (fl) (°C)** | **Active Mud Volume (bbl)** | **Reserve Mud Volume (bbl)** |
| | | | | | | |
| **Cum Mud Added (bbl)** | **Mud Lost to Surface (bbl)** | **Mud Lost to Hole (bbl)** | **Avg Background Gas (%)** | **Avg Drill Gas (%)** | **K** | **n** |
| | | | | | | |

### Marine Conditions

| Swell Height (m) | Visibility (km) | Wind Direction (°) | Wind Speed (knots) | Barometric Pressure (mmHg) | Temperature - High (°C) |
|---|---|---|---|---|---|
| 0.50 | 5. | 337 | 3.0 | 1,011.0 | 24.0 |

### Support Vessels

| Vessel Name | Type | Arrival Date | Depart | Note |
|---|---|---|---|---|
| DP3 | Supply Boat | 2/23/2014 23:30 | | Received 30 Mt FO from Osam Condor, Stby at anchorage. |
| C-Mariner | Supply Boat | 3/21/2014 01:30 | | Delivered deck cargo, transferred Benonite-34Mt,transferring D/Water. |
| Osam Condor | Supply Boat | 3/20/2014 15:45 | | Delivered 30 Mt FO to DP-3, Standby at Anchorage. |

### Mud Additives

| Description | Consumed | Cum Rec | On Loc |
|---|---|---|---|
| Bentonite Bulk | 2.0 | 105.0 | 62.5 |
| Caustic Soda | 1.0 | 427.0 | 235.0 |

### Bulk Amounts

| Supply Item Description | Consumed | Received | Cum On Loc |
|---|---|---|---|
| Potable Water | 22.0 | 22.0 | 102.0 |
| Drill Water | 26.0 | 0.0 | 333.0 |
| Cement G | 0.0 | 0.0 | 183.0 |
| Bentonite Bulk | 2.0 | 34.0 | 88.5 |
| Diesel Fuel | 7.0 | 0.0 | 152.0 |
| Barite BB | 0.0 | 0.0 | 0.0 |
| Barite Bulk | 0.0 | 0.0 | 153.0 |
| Bentonite BB | 0.0 | 0.0 | 22.0 |

### Bit/BHA Information

| BHA No. | Size (in) | Make | Model | Serial Number | Bit Type | IADC Codes | Nozzles (/32") |
|---|---|---|---|---|---|---|---|
| 3 | 24 | Varel | ETR14J | 1250452 | InserT Bit | 435 | 24/24/24/20 |
| **TFA (in²)** | **Drilling Time (hrs)** | **Bit Revs** | **Depth In (mKB)** | **Depth Out (mKB)** | **Depth Drilled (m)** | **BHA ROP (m/hr)** | **IADC Bit Dull** |
| 1.60 | 64.92 | 230,247.00 | 491.00 | 875.00 | 384.00 | 5.9 | 1-1-WT-G-E-I-NO-TD |
| **Flow Rate (min) (gpm)** | **Flow Rate (max) (gpm)** | **SPP (min) (psi)** | **SPP (max) (psi)** | **WOB (min) (1000lbf)** | **WOB (max) (1000lbf)** | **RPM (min) (rpm)** | **RPM (max) (rpm)** |
| 850 | 1,100 | 1,250.0 | 2,100.0 | 22 | 40 | 40 | 80 |
| **Circulating Time (hrs)** | **Rotating Time (hrs)** | **% Rotating Time (%)** | **% Sliding Time (%)** | **PU HL (max) (1000lbf)** | **Rot HL (max) (1000lbf)** | **SO HL (min) (1000lbf)** | **String Wt (1000lbf)** |
| 86.42 | 64.92 | 100.00 | 0.00 | 280 | 256 | 219 | |

# Daily Drilling Report
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### Drilling Parameters

| Depth Start (mKB) | Depth End (mKB) | Drilling Time (hrs) | Drilling Torque | Flow Rate (gpm) | RPM (rpm) | SPP (psi) | WOB (1000lbf) | Drill Str Wt (1000lbf) |
|---|---|---|---|---|---|---|---|---|
| 875.00 | 875.00 | | | | | | | |

### Drill String Components

| Item Description | OD (in) | ID (in) | Len (m) | Cum Len (m) |
|---|---|---|---|---|
| 13 Jts 5 7/8" HWDP | 5 7/8 | 4.250 | 120.97 | 256.31 |
| X/O | 7 | 2.810 | 0.79 | 135.34 |
| 2 Jts 8 1/4" DC | 8 1/4 | 2.875 | 18.75 | 134.55 |
| 8" Cougar DA Jar | 8 | 3.500 | 5.84 | 115.80 |
| 5 Jts 8 1/4" DC | 8 1/4 | 2.875 | 47.24 | 109.96 |
| X/O | 9 5/8 | 3.000 | 1.22 | 62.72 |
| 3 Jts 9 1/2" DC | 9 1/2 | 3.000 | 28.29 | 61.50 |
| 17 1/2" Stabilizer | 17 1/2 | 3.000 | 2.47 | 33.21 |
| 9 1/2" NMDC | 9 1/2 | 4.000 | 9.45 | 30.74 |
| 9 1/2" TM Sub (MWD & Gamma probes inside) | 9 1/2 | 3.125 | 2.15 | 21.29 |
| 9 1/2" NMDC | 9 1/2 | 4.000 | 9.45 | 19.14 |
| 9 1/2" Restrictor sub (32/32" restrictor) | 9 1/2 | 4.250 | 0.91 | 9.69 |
| 23" Stabilizer | 23 | 3.000 | 2.86 | 8.78 |
| 9 1/2" Float Sub (Float valve inside) | 9 1/2 | | 0.79 | 5.92 |
| 9 1/8" Shock Sub | 10 1/8 | 2.625 | 4.51 | 5.13 |
| 24" Insert Bit | 24 | | 0.62 | 0.62 |

### Well Control

| No. | Depth (mKB) | Stroke (in) | Strokes (spm) | Pres (psi) | Eff (%) | Q (gpm) |
|---|---|---|---|---|---|---|
| | | | | | | |

| Test Date | Last Casing String Run | Test Type | Depth (mKB) | TVD (mKB) | Applied Surface Pressure (psi) |
|---|---|---|---|---|---|
| 4/8/2014 | 18 5/8" Shoe, 870.00mKB | Formation Integrity | 870.00 | 843.25 | 400.0 |

| Applied Surface Pressure (psi) | Volume Pumped (bbl) | Current MAASP | Kick Tolerance | Dens Fluid (lb/gal) | Leak Off Pressure (psi) |
|---|---|---|---|---|---|
| 400.0 | 8.0 | 400.00 | | 11.48 | 1,650.4 |

### Geology Information

| Formation Name | Prog Top MD (mKB) | Final Top MD (mKB) | Final Top TVD (mKB) |
|---|---|---|---|
| Lower Dariyan | | 1,183.50 | 1,132.88 |
| Upper Gadvan | | 1,209.00 | 1,156.52 |
| Khalij | | 1,231.00 | 1,176.98 |
| Lower Gadvan | | 1,247.00 | 1,191.88 |
| Fahliyan | | 1,286.50 | 1,228.82 |

### General Remarks

**Comment**

>>> Requirements: (Shock Sub) + (300 MT DW) + (PDF MR# 7).+ (100 MT FO).

>>> Sent: Nil.

>>> Received: Nil.